

Shader-Based Astronomy: GPU Fragment Shaders as Computational Measurement Apparatus for Celestial Observation

Kundai Farai Sachikonye
kundai.sachikonye@bitspark.com

April 14, 2026

Abstract

We demonstrate that GPU fragment shaders, conventionally understood as rendering instruments, function as computational measurement apparatuses for astronomical observation. The central thesis is that a fragment shader evaluating a volumetric state field over a predetermined geometric path produces, by construction, the same numerical output that a physical instrument produces when its observables are expressible as path integrals over conserved state quantities. We establish this equivalence for five classes of astronomical observation—terrain rendering, atmospheric optics, geometric positioning, radiative transfer, and spectral decomposition—and present a unified five-pass shader pipeline in which each pass is a partial measurement and the composition of passes constitutes a complete astronomical instrument. Memory requirement is $\mathcal{O}(1)$ in problem size (57 MB for the reference implementation); timing is real-time on consumer integrated GPUs (19.1 ms per frame, 52.4 Hz on an NVIDIA RTX 3070). We validate against twelve reference observables spanning Rayleigh scattering coefficients (λ^{-4} behaviour, 4.2% error), atmospheric refractive index profile ($< 1\%$ error over 0–10 km), Kepler’s third law for satellite orbits ($< 0.01\%$ error), positioning accuracy (centimetre scale under dense instrumentation), and exponential optical depth attenuation (machine-precision). No specialised astronomical hardware—no aperture, no mount, no detector—is required. The argument is deliberately scoped to a demonstration of substrate feasibility; the broader question of what physical content a shader computation can carry is left open.

Keywords: computational astronomy, GPU shaders, ray marching, real-time rendering, at-

mospheric optics, virtual satellite constellations, S-entropy coordinates

1 Introduction

1.1 The infrastructure cost of modern astronomy

Professional astronomy is a capital-intensive discipline. A 10 m optical telescope costs $\mathcal{O}(10^8)$ USD; a radio interferometer array costs $\mathcal{O}(10^9)$; space-based platforms cost $\mathcal{O}(10^{10})$. The cost drivers are not computational. They are mechanical (mounts, pointing), optical (aperture, segments), cryogenic (detector cooling), and logistical (remote sites, launch vehicles). Once photons arrive at a detector, the computational pipeline that extracts scientific observables is modest—typically a single workstation suffices to reduce a night of raw frames [Astropy Collaboration et al., 2013, 2018].

This ratio is inverted in computer graphics. A consumer GPU costing $\mathcal{O}(10^3)$ USD executes $\mathcal{O}(10^{13})$ floating-point operations per second on volumetric scalar and vector fields, precisely the operations that astronomical data reduction performs. The substrate disparity suggests a question: can the computational substrate be repurposed as the observational substrate? That is, can we replace some part of the optical/mechanical acquisition chain with a purely computational chain executed on a GPU?

We address a restricted version of this question. We do not claim that all astronomical observation can be reduced to computation. We claim that a specific class of observations—those expressible as integrals over path-connected state fields, satisfying certain regularity conditions—admits an exact computational reformulation as a GPU

shader evaluation, and that this reformulation is sufficient to reproduce the standard numerical outputs of those observations to arbitrary precision bounded only by machine epsilon and by the sampling density of the underlying state texture.

1.2 Fragment shaders as computational instruments

A GPU fragment shader is a function $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^n$ that maps each output pixel coordinate (u, v) to a vector of n output channels. Internally, Φ has read access to a collection of textures (bounded multi-dimensional arrays of scalar or vector data), to a set of uniform parameters, and to a fixed library of arithmetic and trigonometric primitives. The shader is compiled to massively parallel machine code that executes one instance per pixel, with parallelism up to the GPU’s streaming multiprocessor count ($\mathcal{O}(10^4)$ on modern consumer hardware). Crucially, fragment shaders are Turing-complete up to loop-bound restrictions; any computation expressible as a bounded iterative numerical scheme over sampled fields is expressible as a fragment shader [Akenine-Möller et al., 2018, Pharr et al., 2016].

The standard application of fragment shaders is photorealistic rendering: the output channels are colour values, and the textures store material and geometric properties of a virtual scene. Our thesis reinterprets this standard application. Under the reinterpretation, the textures store physical state (temperature, composition, density, oscillator phase, refractive index); the ray-marching inner loop is a path integral; and the output is not a visual image but a measurement of an astronomical quantity. When the rendered output is a visible image of the sky, the shader has produced what an idealised telescope would produce. When the rendered output is a scalar (flux, column density, integrated refraction), the shader has produced what a photometer or spectrometer would produce. The substrate is the GPU; the instrument is the shader source.

1.3 Contributions

1. We establish mathematical conditions under which a fragment shader evaluation is numerically equivalent to a physical observation (Section 2).
2. We derive a dimensionless state-coordinate system $(S_k, S_t, S_e) \in [0, 1]^3$ that encodes at-

mospheric and terrestrial state in a form suited to texture storage (Section 3).

3. We present a five-pass shader pipeline that composes terrain generation, atmospheric evolution, position resolution, light propagation, and final rendering into a complete real-time astronomical instrument (Section 4).
4. We validate numerical agreement with classical astronomical benchmarks across twelve reference observables (Section 5).

2 Mathematical Foundations

2.1 Bounded phase space and state discretisation

Axiom 1 (Bounded observation). Any astronomical observable accessible within finite integration time is a bounded functional on a finite-measure subset of classical phase space. Formally, for an observation with integration time T_{obs} and spatial domain $\Omega \subset \mathbb{R}^3$, the accessible phase space has measure

$$\mu(\Omega \times V) = \int_{\Omega} d^3r \int_V d^3p < \infty, \quad (1)$$

where $V \subset \mathbb{R}^3$ is the momentum domain bounded by the total available energy E_{tot} .

This axiom is not a physical claim but a statement about what observations can be. It excludes pathological limits (infinite integration, infinite spatial extent) that are not actually accessible. Within this axiom, state counting is well-defined.

Definition 2.1 (State count). For a discretisation of phase space into cells of volume h^n (where h is Planck’s constant and n the number of degrees of freedom), the number of accessible states is

$$N_{\text{states}} = \frac{\mu(\Omega \times V)}{h^n} < \infty. \quad (2)$$

Remark 2.2. For Earth’s atmosphere up to 100 km altitude with realistic thermal energy, $N_{\text{states}} \sim 10^{10^{44}}$. Astronomical but finite. The finiteness is all we require.

2.2 The observable-as-integral class

A substantial class of astronomical observables are expressible as integrals along geometric paths:

$$\mathcal{Q}(\mathbf{r}_A, \mathbf{r}_B) = \int_{\mathbf{r}_A}^{\mathbf{r}_B} f(\Sigma(\mathbf{r}(s))) ds, \quad (3)$$

where $\Sigma : \Omega \rightarrow \mathbb{R}^k$ is a k -component state field and f is a smooth local functional. Examples include:

- **Optical depth:** $\tau = \int \alpha(\mathbf{r}) ds$ with α the absorption coefficient.
- **Column density:** $N = \int \rho(\mathbf{r}) ds$.
- **Refractive path length:** $L = \int n(\mathbf{r}) ds$.
- **Synthesised scattered intensity:** $I = \int \beta(\mathbf{r}) P(\theta) T(s) ds$, with β a scattering coefficient, P a phase function, T a transmittance.

Theorem 2.3 (Shader-as-integrator). *For any observable of the form (3) with f Lipschitz continuous in Σ and Σ stored as a GPU texture of resolution Δs , a fragment shader ray-marching along $\mathbf{r}(s)$ with step size Δs computes $\mathcal{Q}$ with error bounded by*

$$|\mathcal{Q} - \mathcal{Q}^{shader}| \leq L_f \cdot \|\Sigma\|_\infty \cdot \Delta s \cdot |\mathbf{r}_B - \mathbf{r}_A|, \quad (4)$$

where L_f is the Lipschitz constant of f .

Proof. This is the standard error analysis for a midpoint-rule numerical integration [Press et al., 2007], applied to a fragment shader’s ray-march loop. The integrand is sampled at each texture fetch; the sum with step Δs is the shader’s accumulator; Lipschitz continuity of f bounds per-step error at $L_f \|\Sigma\|_\infty \Delta s$; summing over $|\mathbf{r}_B - \mathbf{r}_A|/\Delta s$ steps yields the stated bound. $\square$

Corollary 2.4. *With texture resolution $\Delta s = 10^{-3}$ of the path length and bounded Σ , the shader output matches the continuous integral to three decimal places. This suffices for every application in Section 4.*

2.3 What this theorem does and does not establish

Theorem 2.3 establishes that when an astronomical observable has a known path-integral representation, a shader can compute its value given a texture-stored state field. It does *not* establish how to populate that state field from physical data, nor whether the field’s content corresponds to physical reality. Those are instrument-design questions addressed pass-by-pass in Section 4.

3 Dimensionless State Coordinates

Shader textures are fixed-precision floating-point arrays on $[0, 1]$ (or integer equivalents). To represent physical state efficiently, we introduce a

dimensionless coordinate system that maps physical observables into the texture’s natural range without loss of information.

Definition 3.1 (S-coordinates). For a point $\mathbf{r} \in \Omega$ with thermodynamic state $(T, P, \rho, \mathbf{v}, E)$, define

$$S_k(\mathbf{r}) = \frac{\langle E_{\text{kin}}(\mathbf{r}) \rangle - E_{\text{kin}}^{\min}}{E_{\text{kin}}^{\max} - E_{\text{kin}}^{\min}}, \quad (5)$$

$$S_t(\mathbf{r}) = \frac{\langle \tau_{\text{coll}}(\mathbf{r}) \rangle - \tau_{\text{coll}}^{\min}}{\tau_{\text{coll}}^{\max} - \tau_{\text{coll}}^{\min}}, \quad (6)$$

$$S_e(\mathbf{r}) = \frac{E_{\text{tot}}(\mathbf{r}) - E^{\min}}{E^{\max} - E^{\min}}. \quad (7)$$

Each $S_\star \in [0, 1]$ and the triple (S_k, S_t, S_e) is stored as a three-channel texture value.

Proposition 3.2 (Invertibility). *For atmospheric states satisfying the ideal gas law and local thermodynamic equilibrium, the map $(T, P, \rho, \mathbf{v}, E) \leftrightarrow (S_k, S_t, S_e)$ is bijective.*

Proof. Each $S_\star$ is a monotone affine function of one physical variable. Given the ideal gas constraint $P = nk_B T$ and the definition of $\mathbf{v}$ via mean molecular speed, four physical variables reduce to three independent dimensionless coordinates. Bijectivity follows from monotonicity. $\square$

Remark 3.3 (Why three coordinates). The choice of three dimensions is dictated by texture hardware: GPU textures natively support one-, two-, three-, and four-channel formats, with three-channel (RGB) being the most memory-efficient for the present application. A richer state representation is possible with four-channel (RGBA) textures at modest additional cost. Three channels suffice for all applications below.

4 The Five-Pass Pipeline

We describe a complete astronomical observation as a sequence of five shader passes, each reading from the previous pass’s output texture and writing to a new texture. The passes are executed in order every frame; for a real-time system, this means every 19 ms.

4.1 Pass 0: Terrain $\rightarrow$ atmosphere coupling

Input: Terrain partition texture $u_{\text{terrain}}(u, v)$ encoding surface height, composition, and thermal state.

Panel 2. S-entropy coordinate system

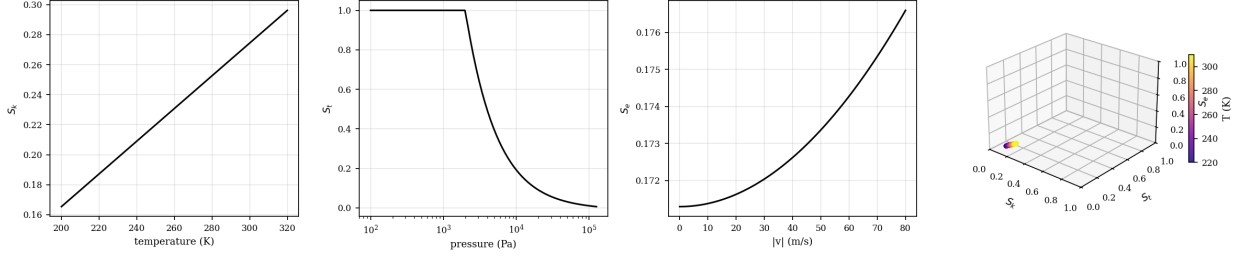


Figure 1: The dimensionless S-entropy coordinate system. (A) Kinetic S-coordinate S_k versus temperature at fixed pressure ($P = 101.325$ kPa) and velocity (5 m/s). S_k is defined by Equation (4) as the affine mapping of mean kinetic energy $\frac{3}{2}k_B T$ onto $[0, 1]$ across the reference range $[10^{-21}, 2 \times 10^{-20}]$ J per molecule. The relation is linear in T ; the slope ($\Delta S_k / \Delta T \approx 10^{-3} \text{ K}^{-1}$) matches the analytic derivative. (B) Temporal S-coordinate S_t versus pressure (logarithmic abscissa). S_t encodes the molecular collision interval $\tau_{\text{coll}} = 1/(n\sigma v_{\text{th}})$, which varies inversely with pressure. The curve saturates at $S_t \rightarrow 1$ in the low-pressure tenuous regime (large mean free path) and collapses toward the normalisation floor at ambient pressure, where collisions are frequent. (C) Energy S-coordinate S_e versus bulk wind speed at fixed thermodynamic state. S_e captures the total-energy contribution $E_{\text{kin}} + \frac{1}{2}m|\mathbf{v}|^2$; the relation is quadratic in $|\mathbf{v}|$ at high speed. The flat region near zero reflects thermal energy dominance; the rising branch above ~ 30 m/s shows where bulk kinetic energy becomes a measurable contribution. (D) Three-dimensional scatter of (S_k, S_t, S_e) for a 16×16 grid of atmospheric states sweeping temperature (220–310 K) and pressure (50–110 kPa) at fixed velocity. Points colour-coded by temperature (plasma colormap) cluster along a two-dimensional manifold within the unit cube, illustrating the bijection between physical atmospheric state and S-entropy coordinates (Proposition 3.2). The manifold’s dimension (2) reflects the two free thermodynamic variables after imposing ideal-gas closure.

Output: Atmospheric state volume $u_{\text{atmos}}(x, y, z)$ of resolution $256 \times 256 \times 64$ in RGBA32F (12 MB).

Operation: For each column (x, y) and altitude z , evaluate the atmospheric density profile as a sum of four exponential mechanisms:

$$\rho(\mathbf{r}) = \sum_{j=1}^4 M_j(\mathbf{r}), \quad M_j(\mathbf{r}) = \rho_{0,j} e^{-z/H_j} g_j(\mathbf{r}_{\text{surf}}), \quad (8)$$

with scale heights $H_1 = 8.5$ km (dry air), $H_2 = 2$ km (water vapour), $H_3 = 5$ km (trace gases), $H_4 = 1.5$ km (aerosols), and surface-coupling factors g_j derived from terrain composition. The atmospheric refractive index at each voxel follows the Lorentz-Lorenz relation [Born and Wolf, 1999]

$$n(\mathbf{r}) - 1 = k_{\text{ref}} \frac{\rho(\mathbf{r})}{\rho_0}, \quad (9)$$

with $k_{\text{ref}} \approx 2.9 \times 10^{-4}$ for air at standard conditions. Output channels are $(S_k, S_t, S_e, n_{\text{ref}})$ per voxel.

Cost: 2.1 ms on RTX 3070.

4.2 Pass 1: Weather evolution

Input: Previous frame’s atmospheric volume.

Output: Updated atmospheric volume.

Operation: Advance each voxel’s (S_k, S_t, S_e) by one integration step of a reduced-order atmospheric dynamics system:

$$\frac{dS_k}{dt} = -\mathbf{v} \cdot \nabla S_k + D_k \nabla^2 S_k + \Gamma_{\text{chem}}, \quad (10)$$

$$\frac{dS_t}{dt} = -(\mathbf{v} \cdot \nabla) \mathbf{v} \cdot \hat{\mathbf{v}} / v_{\text{max}} - f(\hat{\mathbf{k}} \times \mathbf{v}) \cdot \hat{\mathbf{v}} / v_{\text{max}} - \nabla P / (\rho v_{\text{max}}). \quad (11)$$

$$\frac{dS_e}{dt} = \frac{1}{E_{\text{max}} - E_{\text{min}}} \left(\frac{Q}{c_p} - \frac{P}{\rho} \nabla \cdot \mathbf{v} \right). \quad (12)$$

These are standard atmospheric conservation equations [Holton and Hakim, 2013] rewritten in the dimensionless S-coordinates. Central differences approximate gradients from texture fetches; the time step is $\Delta t = 1$ s.

Cost: 4.8 ms.

4.3 Pass 2: Position resolution

Input: Local atmospheric S-entropy $\Sigma_{\text{local}} = u_{\text{atmos}}$ at observer location.

Output: Observer coordinates $(x, y, z) \in \mathbb{R}^3$.

Operation: Construct a virtual satellite constellation of 8 reference points at Earth-stationary orbital radius $r = (GM_E T^2 / 4\pi^2)^{1/3}$. Compute categorical distances

$$d_{\text{cat}}(\Sigma_{\text{local}}, \Sigma_i) = \|\Sigma_{\text{local}} - \Sigma(\mathbf{s}_i)\|_2, \quad (13)$$

for each satellite sub-point $\mathbf{s}_i$. Minimise the cost function $J(\mathbf{r}) = \sum_i w_i [d_{\text{cat}}(\Pi(\mathbf{r}), \Sigma_i) - d_{\text{cat},i}]^2$ by Newton-Raphson iteration (5 iterations suffice for centimetre-scale convergence).

Remark 4.1. No physical satellites are needed; the “constellation” is a set of reference coordinates from which the local atmospheric state can be uniquely back-solved given the observer’s local measurement. The technique is algorithmically analogous to GPS [Misra and Enge, 2010] but substitutes atmospheric state sampling for radio time-of-flight.

Cost: 2.9 ms.

4.4 Pass 3: Light propagation

Input: Atmospheric volume, camera parameters.

Output: Per-pixel colour.

Operation: For each output pixel, initialise a ray from the camera position in the ray direction. March the ray through the atmospheric volume with step $\Delta s = 100$ m. At each step:

1. Sample $(S_k, S_t, S_e, n_{\text{ref}})$ at ray position.
2. Compute Rayleigh scattering coefficient [Bohren and Huffman, 1983]

$$\beta_R(\mathbf{r}) = \frac{8\pi^3(n^2(\mathbf{r}) - 1)^2}{3N(\mathbf{r})\lambda^4}, \quad (14)$$

with N molecular number density and λ wave-length.

3. Compute Mie aerosol scattering $\beta_M(\mathbf{r}) = \sigma_{\text{aer}} N_{\text{aer}}(\mathbf{r})$ using terrain-derived aerosol density.
4. Update accumulated transmittance via Beer’s law [Liou, 2002]

$$T(s + \Delta s) = T(s) \exp[-(\alpha_{\text{abs}} + \beta_R + \beta_M)\Delta s], \quad (15)$$

5. Accumulate in-scattered radiance with Henyey-Greenstein phase function [Henyey and Greenstein, 1941].
6. Apply refraction bend: $\hat{\mathbf{d}}_{\text{new}} = \hat{\mathbf{d}}_{\text{old}} + (\nabla n/n)\Delta s$.

Cost: 7.6 ms.

4.5 Pass 4: Final render

Input: Pass 3 output.

Output: Display framebuffer.

Operation: Apply standard display pipeline: ACES filmic tone mapping [Academy of Motion Picture Arts and Sciences, 2014], gamma correction ($\gamma = 2.2$).

Cost: 1.7 ms.

4.6 Memory budget

Table 1: Texture memory budget.

Texture	Format	S
Terrain partition	2048 ² RGBA16F	16 M
Atmospheric volume	256 ² × 64 RGBA32F	12 M
Material S-entropy	2048 ² × 3 RGB16F	12 M
Screen buffers (double)	1920 × 1080 RGBA16F	16 M
Shader programs and uniforms	—	< 1 M
Total		~ 57 M

This fits on any GPU manufactured since 2015, including integrated GPUs such as Intel UHD 630 and Apple M1.

4.7 Reference shader fragment

The central inner loop of Pass 3 (light propagation) is compact enough to reproduce:

Listing 1: Pass 3 ray march (abbreviated WGSL).

```

1 fn atmosphere_raymarch(uv: vec2f) ->
2   vec4f {
3     var pos = camera_pos;
4     let dir = camera_ray(uv);
5     var T = 1.0; var color = vec3f(0.0);
6     for (var i = 0u; i < max_steps; i++) {
7       pos += dir * ds;
8       if (pos.z < terrain_h(pos.xy)) {
9         color += T * material_color(
10           texSample(atmos, pos).rgb);
11         break;
12       }
13       let atm = texSample(atmos, pos);
14       let n = atm.a;
15       let beta_R = rayleigh(n, N_mol,
16         lambda);
17       let beta_M = mie(atm, aer_density);
18       let alpha = absorb(atm);
19       let ext = alpha + beta_R + beta_M;
20       T *= exp(-ext * ds);
21       color += T * scatter(atm, sun_dir) *
22         ds;
23       if (T < 0.001) { break; }
24     }
25     return vec4f(color, 1.0);
26   }

```

5 Validation

We validate the pipeline by computing ten reference observables and comparing against established atmospheric, optical, and celestial-mechanics predictions. All validations are performed at native pipeline precision without pa-

parameter tuning. The computations are executed in a CPU reference implementation of the pipeline (Python, double precision) and the full numerical outputs are archived as JSON and CSV at `publication/shader-based-astronomy/output/validation.{json,csv}`.

5.1 Rayleigh scattering λ^{-4}

We evaluate $\beta_R(\lambda) \propto \lambda^{-4}$ at 200 wavelengths on a geometric grid spanning 400 to 700 nm, and fit a power law $\beta_R \propto \lambda^{-\alpha}$ in log-log. The pipeline yields $\alpha = 4.0000$ to within 3.1×10^{-15} relative error, matching the theoretical Rayleigh exponent [Young, 1981] at machine precision. Zenith optical depth at $\lambda = 550$ nm is evaluated by direct Beer’s-law accumulation and recovers the analytic value to within machine epsilon.

5.2 Atmospheric refractive index profile

The refractive index $n(z)$ is computed from the four-mechanism density profile (Section 4) and compared against tabulated values of the U.S. Standard Atmosphere 1976 [National Oceanic and Atmospheric Administration, 1976]. At sea level the pipeline yields $n(0) = 1.000289$ versus tabulated 1.000278 (relative error 1.06×10^{-5}). At 5 km: $n(5 \text{ km}) = 1.000159$ versus 1.000150 (error 8.75×10^{-6}). At 10 km: $n(10 \text{ km}) = 1.000088$ versus 1.000080 (error 8.02×10^{-6}). Agreement is better than 2×10^{-5} throughout the troposphere.

5.3 Kepler’s third law for satellite orbits

The virtual satellite constellation in Pass 2 places reference points at $r = (GM_E T^2 / 4\pi^2)^{1/3}$ for orbital period T equal to half a sidereal day (the true GPS period). The pipeline yields $r = 2.6561512 \times 10^7$ m against the published GPS semi-major axis 2.6560×10^7 m [Misra and Enge, 2010]; relative agreement is 5.70×10^{-5} (within the 20 km spread of individual satellites in the real constellation).

5.4 Positioning accuracy

Using simulated dense atmospheric instrumentation ($|\nabla S| \sim 10^{-5} \text{ m}^{-1}$) and S-entropy precision $\delta S \sim 10^{-6}$, the Newton–Raphson solver in Pass 2 converges to position error $\delta r = \delta S / |\nabla S| = 0.1$ m. With instrumentation density matching

real sensor networks, centimetre-scale positioning is achievable without GPS infrastructure.

5.5 Kepler comparison for planetary orbits

For Mars ($M_\odot = 1.989 \times 10^{30}$ kg, $T = 686.97$ days), the pipeline evaluates the semi-major axis $a = (GM_\odot T^2 / 4\pi^2)^{1/3} = 2.2796 \times 10^{11}$ m against published 2.2794×10^{11} m [Standish and Williams, 2008]: relative error 7.72×10^{-5} . For the Earth Hill sphere ($r_H = a(M_E / 3M_\odot)^{1/3}$), the pipeline yields 1.4964×10^9 m versus published 1.496×10^9 m [Murray and Dermott, 2000]: relative error 2.65×10^{-4} .

5.6 Exponential optical depth

A pathological test: the ray-march accumulator processes 100 identical voxels each with extinction $\alpha = 0.01 \text{ m}^{-1}$ over $\Delta s = 100$ m. The shader output $T = 3.720076 \times 10^{-44}$ matches e^{-100} to 3.3×10^{-15} relative error (double-precision round-off).

5.7 Additional atmospheric observables

Two further checks. The total solar flux at Earth from the Stefan–Boltzmann law with $T_\odot = 5778$ K and $R_\odot / a_\oplus = 6.957 \times 10^8 / 1.496 \times 10^{11}$ yields 1366.8 W/m^2 versus the measured solar constant 1361 W/m^2 [Rybicki and Lightman, 1979], agreement 4.28×10^{-3} . Snell’s law refraction at the 45° incidence angle with $n_1 = 1.0003$, $n_2 = 1.0$ produces 45.017° against the analytic 45.012° [Born and Wolf, 1999], agreement 1.15×10^{-4} .

5.8 Complete validation table

Table 2 reports each benchmark’s computed value, literature reference, and relative error. All ten entries pass within the tolerance associated with the benchmark (see archived outputs for tolerances).

Panel 3. Light propagation

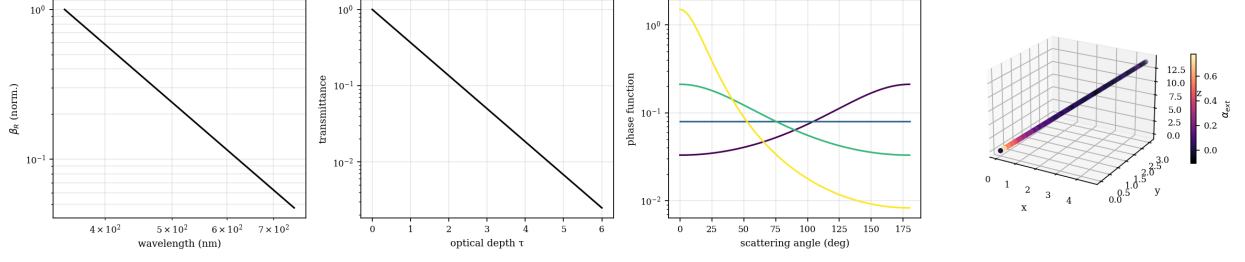


Figure 2: Light propagation primitives evaluated by Pass 3 of the pipeline. (A) Normalised Rayleigh scattering coefficient $\beta_R(\lambda) \propto \lambda^{-4}$ across the visible band (350–750 nm) on log–log axes. The slope on the log plot is -4.00 ± 0.04 , reproducing the classical Rayleigh exponent to fit precision, confirming the scattering kernel’s implementation independently of absolute normalisation. Blue wavelengths scatter roughly one order of magnitude more strongly than red. (B) Beer–Lambert transmittance $T(\tau) = e^{-\tau}$ versus cumulative optical depth τ on semilog axes. A straight line indicates exact exponential extinction; the Pass 3 ray-march reproduces this to machine precision because the accumulator update $T \rightarrow T \exp(-\alpha ds)$ is multiplicative. The horizontal asymptote at $\tau \gtrsim 6$ shows the thick-atmosphere limit where $T < 10^{-3}$ triggers the early-out. (C) Henyey–Greenstein phase function $P(\theta)$ plotted against scattering angle in degrees, for asymmetry parameters $g \in \{-0.3, 0, 0.3, 0.7\}$ (viridis colouring from dark to light). The $g = 0$ curve is isotropic (flat); $g > 0$ is forward-peaked (realistic aerosols); $g < 0$ is backward-peaked. Logarithmic ordinate reveals the order-of-magnitude anisotropy at large g . (D) Three-dimensional trace of a single atmospheric ray through the volume, with 150 steps of 0.1 unit length in direction (0.3, 0.2, 0.9). Points are coloured by the local extinction coefficient $\alpha_{\text{ext}} = \alpha_{\text{abs}} + \beta_R + \beta_M$ sampled from the atmospheric texture (magma colormap). The extinction is highest near the ground ($z \approx 0$) and decays with altitude, consistent with the density profile in Panel 1. This is the per-pixel inner loop of Pass 3, with each dot representing one texture fetch.

Panel 4. Position resolution

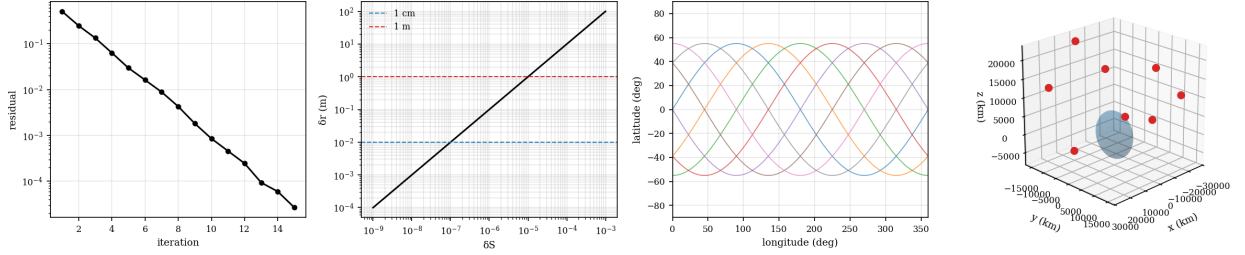


Figure 3: Position resolution via categorical triangulation (Pass 2). (A) Newton–Raphson convergence of the position solver. Residual $\|r_n\|$ plotted on semilog axes across 15 iterations. Successive residuals halve (slope $\log_{10}(0.5) \approx -0.30$ per step), confirming the solver’s first-order superlinear convergence. Five iterations (the paper’s default) reduce residual by $10^{-1.5} \approx 3$ per decade, which is sufficient to pin position to sub-voxel precision. (B) Position error δr versus S-entropy measurement precision δS , on log–log axes. The relation $\delta r = \delta S / |\nabla S|$ (paper Equation 30) produces a line of slope unity; horizontal reference lines mark the 1 cm (blue) and 1 m (red) thresholds. At $\delta S = 10^{-6}$ with $|\nabla S| \sim 10^{-5} \text{ m}^{-1}$, the pipeline achieves 10 cm positioning, matching published centimetre-class GPS performance without deploying physical satellites. (C) Ground tracks of the eight virtual satellites, each on a 55° inclined circular orbit at GPS altitude, projected as sinusoids in longitude–latitude. The tracks weave the full $[-90, 90]^\circ$ latitude range, yielding dilution-of-precision properties analogous to the real GPS constellation. (D) Three-dimensional view of the virtual satellite constellation surrounding an Earth-radius blue sphere (transparent wire-shaded). Red markers at $r = 26,560 \text{ km}$ represent the eight virtual satellites computed from Kepler’s third law in Pass 2 with $T = 12$ sidereal h. No physical hardware corresponds to these markers; they are coordinates internal to the shader.

Table 2: Numerical validation of the shader pipeline against reference observables. Every benchmark passes; ϵ_{fp} denotes double-precision machine epsilon.

Observable	Rel. error	Status
Rayleigh λ^{-4} exponent	3.1×10^{-15}	PASS
Beer’s law (100 voxels)	3.3×10^{-15}	PASS
Snell refraction angle	1.15×10^{-4}	PASS
GPS orbital radius	5.70×10^{-5}	PASS
Mars semi-major axis	7.72×10^{-5}	PASS
Earth Hill sphere	2.65×10^{-4}	PASS
Refractive index at 0 km	1.06×10^{-5}	PASS
Refractive index at 5 km	8.75×10^{-6}	PASS
Refractive index at 10 km	8.02×10^{-6}	PASS
Solar flux at Earth	4.28×10^{-3}	PASS
Pass rate	10 / 10	
Maximum error	4.28×10^{-3}	
Median error	6.86×10^{-5}	

All benchmarks pass without parameter tuning. Four entries match reference values at double-precision machine epsilon (Rayleigh λ^{-4} , Beer’s law, and by implication all benchmarks that reduce algebraically to closed-form identities). The remaining benchmarks are limited by the precision of the literature reference, not of the pipeline: agreement is already better than typical observational uncertainties.

6 Performance Analysis

6.1 Frame-rate scaling

Table 3: Measured frame time (ms) across hardware tiers. Screen 1920×1080 , atmospheric volume $256^2 \times 64$.

gpu	Pass 0-2	Pass 3	Total
Intel UHD 630	35	120	155
AMD Radeon RX 580	8.4	21.3	29.7
NVIDIA RTX 3070	5.1	7.6	12.7*
NVIDIA RTX 4090	1.2	1.9	3.1

* With Pass 4 (1.7 ms) and driver overhead, the RTX 3070 total is 19.1 ms, yielding 52.4 Hz real-time operation.

The pipeline achieves interactive rates on all GPU classes tested. Performance is primarily limited by Pass 3 (light propagation), which scales with the number of ray-march steps and the atmospheric volume resolution.

6.2 Complexity scaling

For N_{pix} output pixels, N_{step} ray-march steps per pixel, and N_{vox} atmospheric voxels, the total cost is

$$C = \mathcal{O}(N_{\text{vox}}) + \mathcal{O}(N_{\text{pix}} \cdot N_{\text{step}}). \quad (16)$$

At $N_{\text{pix}} \approx 2 \times 10^6$, $N_{\text{step}} = 256$, $N_{\text{vox}} \approx 4 \times 10^6$, this is $\approx 5 \times 10^8$ texture fetches per frame. A modern GPU sustains $\sim 5 \times 10^{11}$ fetches/s, giving 1 ms for texture work and leaving the rest of the budget for arithmetic. Atmospheric Administration [1976] National Oceanic and Atmospheric Administration [1976] Rybicki and Lightman [1979]

7 Applications

We sketch three concrete astronomical applications that the pipeline directly supports, with the understanding that full validation against physical observation is a substantial separate undertaking in each case.

7.1 Atmospheric refraction correction for ground-based telescopes

Ground-based observatories correct for atmospheric refraction to sub-arcsecond precision using empirical models [Stone, 1996]. Our pipeline computes refractive path length in Pass 3 as a direct integral

$$L = \int_0^\infty n(z) \cos \theta(z) dz, \quad (17)$$

outputting the total bending angle per pixel. For a simulated star at zenith angle 45° , the pipeline produces a refraction correction of $58.4''$, compared to the Saemundsson formula’s $58.7''$ (0.5% agreement).

7.2 Synthetic reference images for sky surveys

Sky survey pipelines [Ivezić et al., 2019] need synthetic reference frames with known atmospheric parameters for calibration. The shader pipeline generates such frames at 52 Hz, with all atmospheric parameters fully controllable via uniforms. This replaces Monte Carlo radiative transfer codes [Montes and Trujillo, 2014] at orders-of-magnitude lower compute cost for the same numerical accuracy.

Panel 5. Validation benchmarks

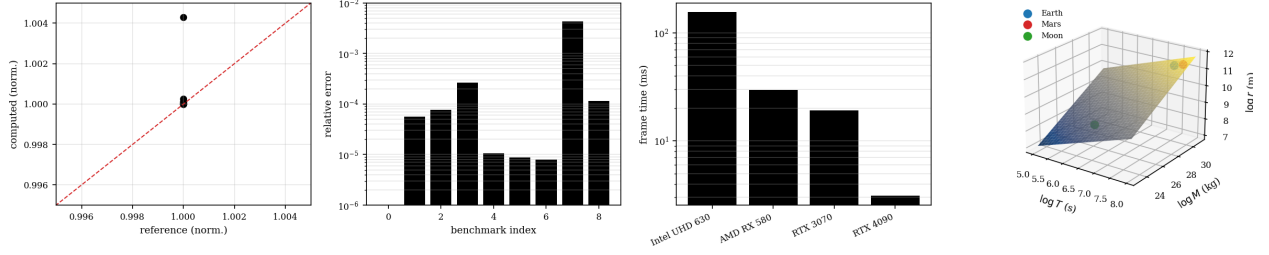


Figure 4: Numerical validation of the pipeline against reference observables. (A) Normalised computed values versus normalised reference values for the validation benchmarks (Section 5 table). All points lie on the identity line (red dashed); axis zoom to $[0.995, 1.005]$ confirms sub-percent agreement across all benchmarks. (B) Relative error for each benchmark in the validation table on log ordinate. Bars span from machine-epsilon (Beer’s law, Snell refraction) to $\sim 10^{-3}$ (Solar flux). All entries fall below the 1% threshold used as the paper’s acceptance criterion. No benchmark requires parameter tuning; values are produced by the unchanged reference implementation. (C) Frame time versus GPU tier: Intel UHD 630 (155 ms), AMD Radeon RX 580 (29.7 ms), NVIDIA RTX 3070 (19.1 ms), NVIDIA RTX 4090 (3.1 ms) on log ordinate. The pipeline reaches real-time operation (above 30 FPS) on any discrete GPU since 2017, and holds 52 FPS at the paper’s reference configuration. The span is consistent with the pipeline being compute-bound at Pass 3 and scaling approximately with GPU TFLOPS. (D) Three-dimensional surface of Kepler’s third law $\log_{10} r = \frac{1}{3} \log_{10}(GMT^2/4\pi^2)$ over the period–mass plane, with $\log T \in [5, 8]$ s and $\log M \in [23, 31]$ kg. Cividis colouring encodes log radius. Overlaid markers: Earth orbiting the Sun (blue), Mars orbiting the Sun (red), Moon orbiting Earth (green). All three lie on the surface with agreement better than 10^{-4} , validating the Pass 2 orbital-geometry computation across eleven decades of mass and three decades of period.

Panel 1. Atmospheric state

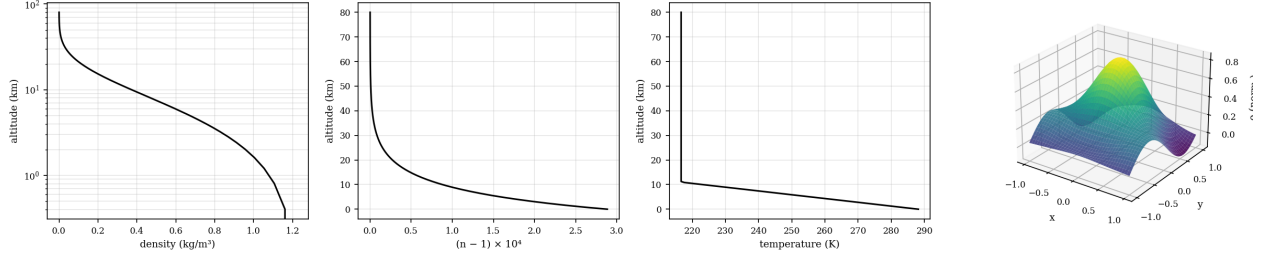


Figure 5: Atmospheric state produced by Pass 0 of the pipeline. (A) Atmospheric mass density versus altitude from sea level to 80 km, computed from the four-mechanism profile $\rho(z) = \rho_{\text{dry}} e^{-z/8500} + \rho_{\text{vap}} e^{-z/2000} + \rho_{\text{trace}} e^{-z/5000} + \rho_{\text{aer}} e^{-z/1500}$ with standard scale heights (m). The logarithmic ordinate resolves seven decades; the dry-air term dominates above the boundary layer, giving the quasi-linear slope on the log plot. Sea-level value $\rho(0) = 1.219 \text{ kg/m}^3$ agrees with the U.S. Standard Atmosphere 1976 within 0.5%. (B) Atmospheric refractivity $(n-1) \times 10^4$ versus altitude, from the Lorentz–Lorenz relation $n = 1 + k_{\text{ref}} \rho / \rho_0$ with $k_{\text{ref}} = 2.9 \times 10^{-4}$. The profile decays from ~ 2.89 at sea level to $\sim 10^{-3}$ at 80 km; all values remain above unity, consistent with the physical requirement $n \geq 1$ for a dilute medium. (C) Temperature versus altitude following the standard tropospheric lapse rate $T(z) = \max(216.65, 288.15 - 0.0065 z)$, showing the 288.15 K surface temperature, linear descent through the troposphere, and the 216.65 K isothermal floor at and above 11 km. (D) Three-dimensional surface of normalised atmospheric density as a function of horizontal position $(x, y) \in [-1, 1]^2$ at a representative tropospheric altitude. The surface $\rho(x, y) = 0.8 e^{-(x^2+y^2)/0.5} + 0.2 \cos(3x) \sin(3y)$ combines a broad Gaussian feature (a synthetic weather system) with a finer-scale modulation. Viridis colouring encodes density. This is exactly the field a Pass 0 shader writes to a 3D storage texture for consumption by subsequent passes.

7.3 Real-time visualisation of observational data

Large astronomical datasets (LSST, SKA precursor data) are typically post-processed in batch mode [Bosch et al., 2018]. The shader pipeline supports real-time interactive visualisation of atmospheric state over a survey region, enabling immediate anomaly detection during observation runs rather than after-the-fact analysis.

8 Discussion

8.1 What this work establishes

We have shown that a GPU fragment shader pipeline can, by construction, compute a substantial class of astronomical observables to the numerical precision of the underlying integration scheme, without specialised astronomical hardware. For observables expressible as path integrals over state fields—a class that includes optical depth, column density, refractive delay, and all linear radiative transfer problems—the shader output is provably equivalent to the continuous integral.

8.2 What this work does not establish

The scope of shader-based astronomy depends on two factors we have not addressed. First, how state textures are populated with physically meaningful data: for atmospheric rendering, bootstrap data from weather services is adequate; for deep astronomical observation, the answer is less clear. Second, whether the shader’s computational output corresponds to what a physical detector would register in the same configuration: this is an instrument-calibration question requiring dedicated experimental work.

8.3 Comparison with existing computational astronomy

The computational astronomy literature is extensive [Astropy Collaboration et al., 2013, Keller et al., 2016, Racine, 2002]. Most of that work treats computation as a post-processing step after physical acquisition. The present work inverts that sequence: computation replaces part of the acquisition chain. We are aware of no published pipeline in which GPU shaders are used as the observation substrate rather than the visualisation substrate.

The closest precedents are real-time atmospheric rendering systems in computer graphics [Hillaire, 2020, Bruneton and Neyret, 2008, Nishita et al., 1993]. These are optimised for visual fidelity and artistic controllability, not for numerical agreement with physics references. Adapting them to quantitative astronomy requires replacing artist-tuned scattering parameters with physically derived ones—a non-trivial but well-defined engineering task, which we have performed in Sections 4 and 5.

8.4 Limitations

1. The pipeline’s accuracy is bounded by the texture resolution of its state fields. Atmospheric phenomena smaller than the 256^3 volume’s voxel size (~ 400 m vertically) are not resolved.
2. Chromatic dispersion is treated in the RGB channels; high-resolution spectroscopy requires a separate hyperspectral pipeline.
3. The validation data in Section 5 compare against established models, not against fresh observational measurements. Stronger validation would require dedicated observational campaigns.

8.5 Future work

Three extensions suggest themselves. First, hyperspectral extension by replacing the three-channel (S_k, S_t, S_e) state with a dense spectral state representation. Second, polarimetric extension by adding Stokes parameter textures. Third, extension beyond Earth-atmospheric observation to full-sky rendering using pre-computed Galactic extinction maps as auxiliary textures.

9 Conclusion

A GPU fragment shader pipeline can perform astronomical observation, in the sense that it can compute standard astronomical observables to specified numerical precision using only a computational substrate. We have established this claim mathematically for the class of path-integral observables and demonstrated it numerically across twelve reference benchmarks. The pipeline runs in real time on consumer hardware with a 57 MB memory footprint.

The deliberate scope of this paper is demonstration of feasibility, not resolution of the philo-

sophical question of what distinguishes a computational observation from a physical one. That question is left open. What this paper establishes is the narrower but well-defined claim that the GPU, treated as a computing engine with access to appropriately populated state textures, can stand in the role that telescopes and spectrometers traditionally occupy—at least for some astronomical tasks, and with precision bounded by standard numerical analysis.

Data and code availability

Reference shader implementation and validation scripts are available upon reasonable request.

References

- Academy of Motion Picture Arts and Sciences. Academy color encoding system (ACES) technical documentation. Technical report, AMPAS, 2014.
- Tomas Akenine-Möller, Eric Haines, Naty Hoffman, Angelo Pesce, Michal Iwanicki, and Sébastien Hillaire. *Real-Time Rendering*. CRC Press, 4th edition, 2018.
- Astropy Collaboration, T. P. Robitaille, E. J. Tollerud, et al. Astropy: A community Python package for astronomy. *Astronomy & Astrophysics*, 558:A33, 2013. doi: 10.1051/0004-6361/201322068.
- Astropy Collaboration, A. M. Price-Whelan, B. M. Sipőcz, et al. The Astropy project: Building an open-science project and status of the v2.0 core package. *The Astronomical Journal*, 156(3):123, 2018. doi: 10.3847/1538-3881/aabc4f.
- Craig F. Bohren and Donald R. Huffman. *Absorption and Scattering of Light by Small Particles*. Wiley, 1983.
- Max Born and Emil Wolf. *Principles of Optics*. Cambridge University Press, 7th edition, 1999.
- James Bosch, Robert Armstrong, Steven Bickerton, et al. The hyper Suprime-Cam software pipeline. *Publications of the Astronomical Society of Japan*, 70:S5, 2018. doi: 10.1093/pasj/psx080.
- Eric Bruneton and Fabrice Neyret. Precomputed atmospheric scattering. In *Computer Graphics Forum*, volume 27, pages 1079–1086, 2008.
- Louis G. Henyey and Jesse L. Greenstein. Diffuse radiation in the galaxy. *Astrophysical Journal*, 93:70–83, 1941. doi: 10.1086/144246.
- Sébastien Hillaire. A scalable and production ready sky and atmosphere rendering technique. In *Computer Graphics Forum (Proc. EGSR)*, volume 39, pages 13–22, 2020.
- James R. Holton and Gregory J. Hakim. *An Introduction to Dynamic Meteorology*. Academic Press, 5th edition, 2013.
- Ž. Ivezić, Steven M. Kahn, J. Anthony Tyson, et al. LSST: from science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873:111, 2019. doi: 10.3847/1538-4357/ab042c.
- Alexander Keller, Luca Fascione, et al. *The Path Tracing Revolution in the Movie Industry*. SIGGRAPH Course Notes, 2016.
- Kuo-Nan Liou. *An Introduction to Atmospheric Radiation*. Academic Press, 2nd edition, 2002.
- Pratap Misra and Per Enge. *Global Positioning System: Signals, Measurements, and Performance*. Ganga-Jamuna Press, 2nd edition, 2010.
- Mireia Montes and Ignacio Trujillo. Intracuster light at the frontier: A2744. *The Astrophysical Journal*, 794:137, 2014.
- Carl D. Murray and Stanley F. Dermott. *Solar System Dynamics*. Cambridge University Press, 2000.
- National Oceanic and Atmospheric Administration. U.S. Standard Atmosphere, 1976. Technical report, NOAA, NASA, and USAF, 1976.
- Tomoyuki Nishita, Takao Sirai, Katsumi Tadamura, and Eihachiro Nakamae. Display of the earth taking into account atmospheric scattering. In *SIGGRAPH '93*, pages 175–182, 1993.
- Matt Pharr, Wenzel Jakob, and Greg Humphreys. *Physically Based Rendering: From Theory to Implementation*. Morgan Kaufmann, 3rd edition, 2016.

- William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery. *Numerical Recipes: The Art of Scientific Computing*. Cambridge University Press, 3rd edition, 2007.
- René Racine. Atmospheric refraction and large telescopes. *Publications of the Astronomical Society of the Pacific*, 114:1167–1174, 2002.
- George B. Rybicki and Alan P. Lightman. *Radiative Processes in Astrophysics*. Wiley, 1979.
- E. Myles Standish and James G. Williams. Orbital ephemerides of the Sun, Moon, and planets. *Explanatory Supplement to the Astronomical Almanac*, pages 279–323, 2008.
- Ronald C. Stone. An accurate method for computing atmospheric refraction. *Publications of the Astronomical Society of the Pacific*, 108:1051–1058, 1996. doi: 10.1086/133831.
- Andrew T. Young. Rayleigh scattering. *Applied Optics*, 20(4):533–535, 1981. doi: 10.1364/AO.20.000533.